SOFTWARE QUALITY ENGINEERING

ASSIGNMENT#01

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Title: Black Box Testing Methods

Black Box Testing:

Black Box Testing (also known as Behavioral Testing) is a software testing method where the internal structure, design, and implementation of the item being tested are not known to the tester.

The tester focuses solely on the inputs and outputs of the software system. It is called "Black Box" because the software is treated like an opaque box; you cannot see how the code works inside, you only care if the system behaves correctly based on the requirements.

Key Testing Techniques:

1. Equivalence Class Partitioning (ECP):

This technique divides the input data of a software unit into partitions of equivalent data from which test cases can be derived.

- The Goal: To minimize the number of test cases to an optimum level while maintaining reasonable coverage.
- How it works: If one value in a partition works, all other values in that same partition are assumed to work. For example, if a text field accepts ages 18–60, you would test one value from the "Invalid Low" range (e.g., 10), one from the "Valid" range (e.g., 25), and one from the "Invalid High" range (e.g., 70).

2. Boundary Value Analysis (BVA)

BVA is closely related to Equivalence Partitioning but focuses specifically on the values at the boundaries.

- The Goal: To identify errors that often occur at the edges of input ranges rather than in the middle.
- How it works: If a system accepts values from 1 to 100, BVA will test the minimum (1), the maximum (100), and values just outside these boundaries (0 and 101).

3. Decision Table Testing

A Decision Table is a systematic way to document complex business rules and logic.

- The Goal: To capture different combinations of inputs and their associated outputs in a matrix format.
- How it works: Each column in the table represents a unique combination of input conditions (causes) and the resulting action (effect). This is particularly useful for systems with many "If-Then" logic paths.

4. State Transition Testing

This technique is used when the system transitions from one state to another based on different inputs.

- The Goal: To verify that the system moves correctly through its lifecycle.
- Example: An ATM machine. It starts at the "Waiting for Card" state. Once a card is inserted, it moves to "Waiting for PIN." If the PIN is correct, it moves to "Account Access." If incorrect three times, it moves to "Card Blocked."

5. Error Guessing

This is an experience-based technique where the tester uses their intuition and previous knowledge to "guess" where errors are most likely to occur.

- The Goal: To find bugs that formal techniques might miss.
- How it works: A tester might purposely enter special characters into a name field or try to submit a form without filling in mandatory fields to see how the system handles it.

Why use Black Box Testing?

- **User Perspective:** It validates the software against the actual requirements and user expectations.
- **Unbiased:** Since the tester doesn't know the code, they aren't influenced by how the developer *meant* for it to work.
- **Early Implementation:** Test cases can be designed as soon as the specifications are finished, even before coding begins.
- **No Coding Knowledge Required:** Testers do not need to be programmers to effectively find functional flaws